Exploratory Data Analysis of YouTube Music Activity

Student: Dat Huynh, Quan Nguyen, Evan Fraustro, Daniil Bondar, Matthew Tindall
Course: COSC 3337 Data Science
Instructor: Dr. Raunak Sarbajna
Date: 03/20/2026

1. Introduction

This report presents a comprehensive exploratory data analysis (EDA) of YouTube watch and search history data. The objective of this analysis is to uncover patterns in user behavior, identify trends in music consumption, and compare engagement across multiple years. The dataset provides valuable insight into how users interact with digital media platforms, particularly in the context of music and entertainment content.

Exploratory data analysis is a crucial step in understanding datasets before applying advanced modeling techniques. In this project, various statistical summaries and visualization techniques are applied to gain insights into user preferences and behavior over time.

2. Dataset Description

The dataset consists of two primary components: watch history and search history. The watch history dataset contains approximately 49,400 entries, while the search history dataset contains 6,912 entries.

Each record includes:
- Title: Represents the name of the video watched. This field is useful for text analysis and identifying content type, such as music, reactions, or entertainment videos.
- Time: A timestamp indicating when the video was watched. This attribute is critical for temporal analysis, allowing extraction of year, month, and day to study trends over time.
- Channel: The content creator or source of the video. This attribute helps identify user preferences and frequently watched creators.
- Additional metadata such as product and activity type

Derived features were also created:
- Year: Extracted from the timestamp, used to compare activity across different years.
- Month: Extracted from the timestamp, used to analyze seasonal trends and monthly patterns.
- Day: Provides finer granularity for time-based analysis, though less significant than year

and month for this study.

These attributes together enable both statistical analysis and visualization of user behavior.

3. Data Preprocessing

Data preprocessing was necessary due to inconsistencies in timestamp formats. Some entries included milliseconds, while others did not. To resolve this, timestamps were parsed using ISO8601 format.

Additional preprocessing steps included:
- Extracting channel names from nested fields
- Handling missing or malformed entries
- Cleaning text data by removing non-informative words such as 'watched', 'YouTube', and 'video'

These steps ensured that the dataset was consistent and suitable for analysis.

4. Summary Statistics

The dataset contains a total of 49,400 watch entries. The distribution of activity across years is as follows:
- 2025: Highest activity (~34,500 entries)
- 2026: Moderate activity (~7,000 entries, partial year)
- 2023 and 2024: Lower activity (~3,500 entries each)

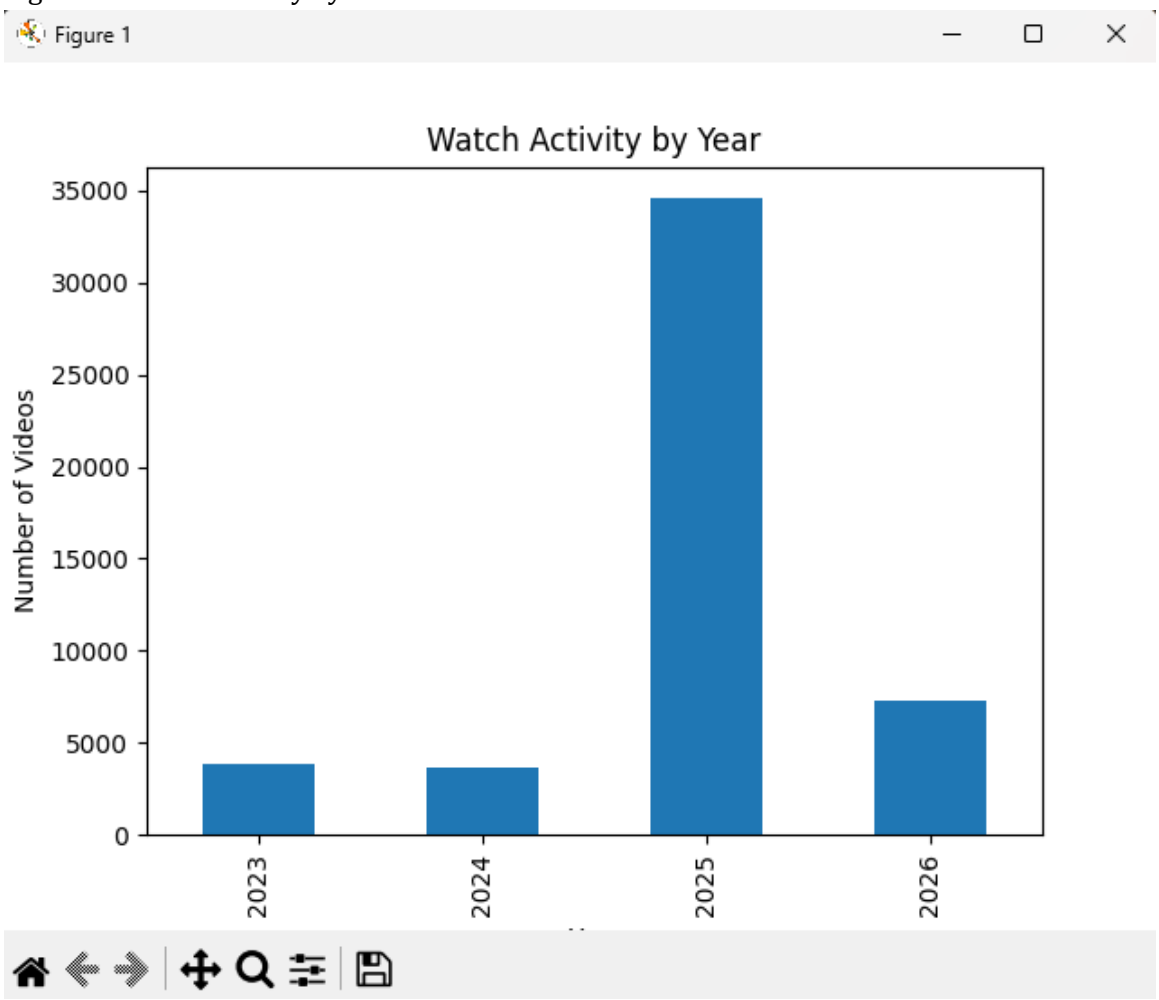
Top channels include:
- Sơn Tùng M-TP
- SOOBIN
- B Ray
- Khảm Vlogs
- Acoustic Bar

These statistics indicate a strong preference for music-related content.

5. Visual Analysis

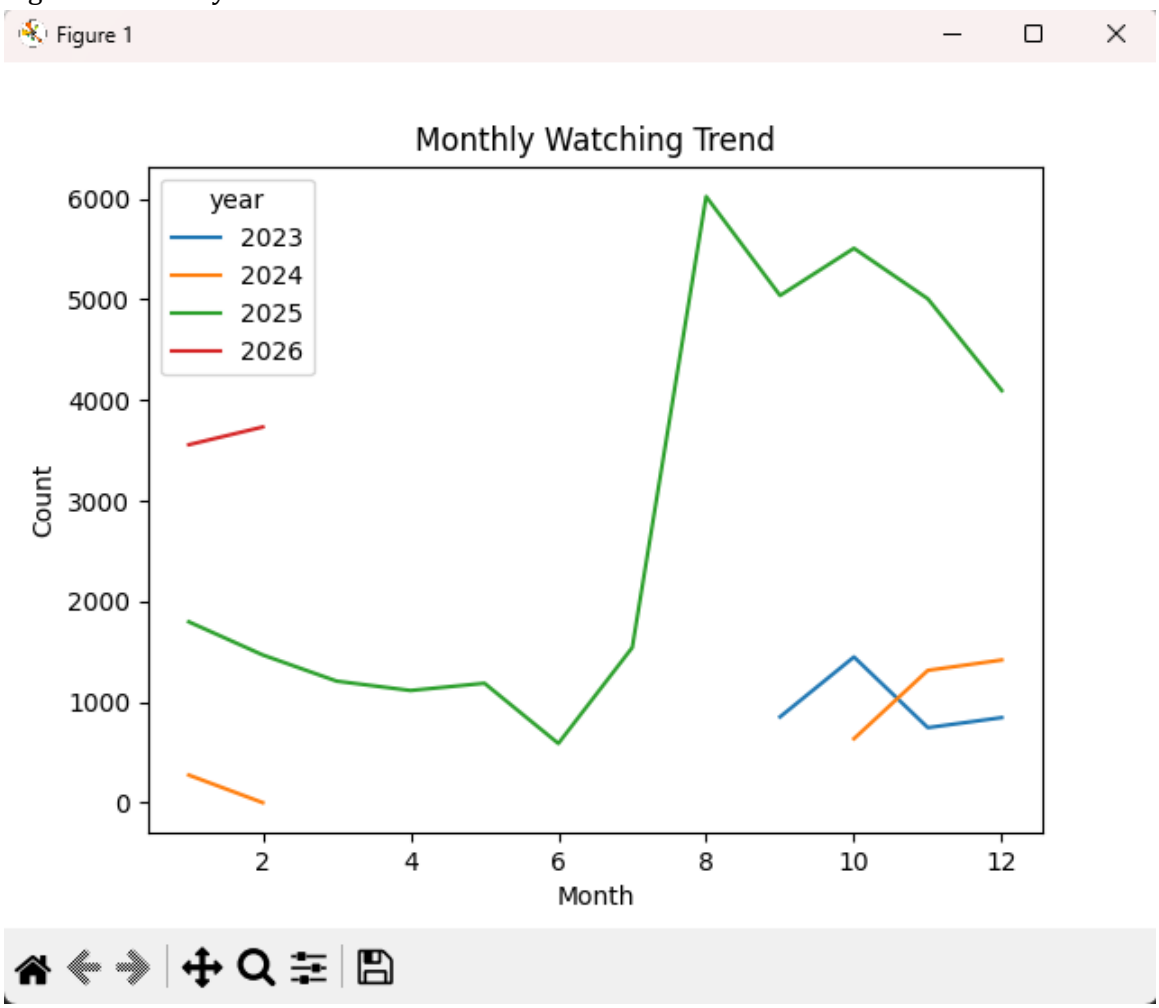
This section presents various visualizations used to analyze the dataset. These include histograms, line charts, box plots, and word clouds.

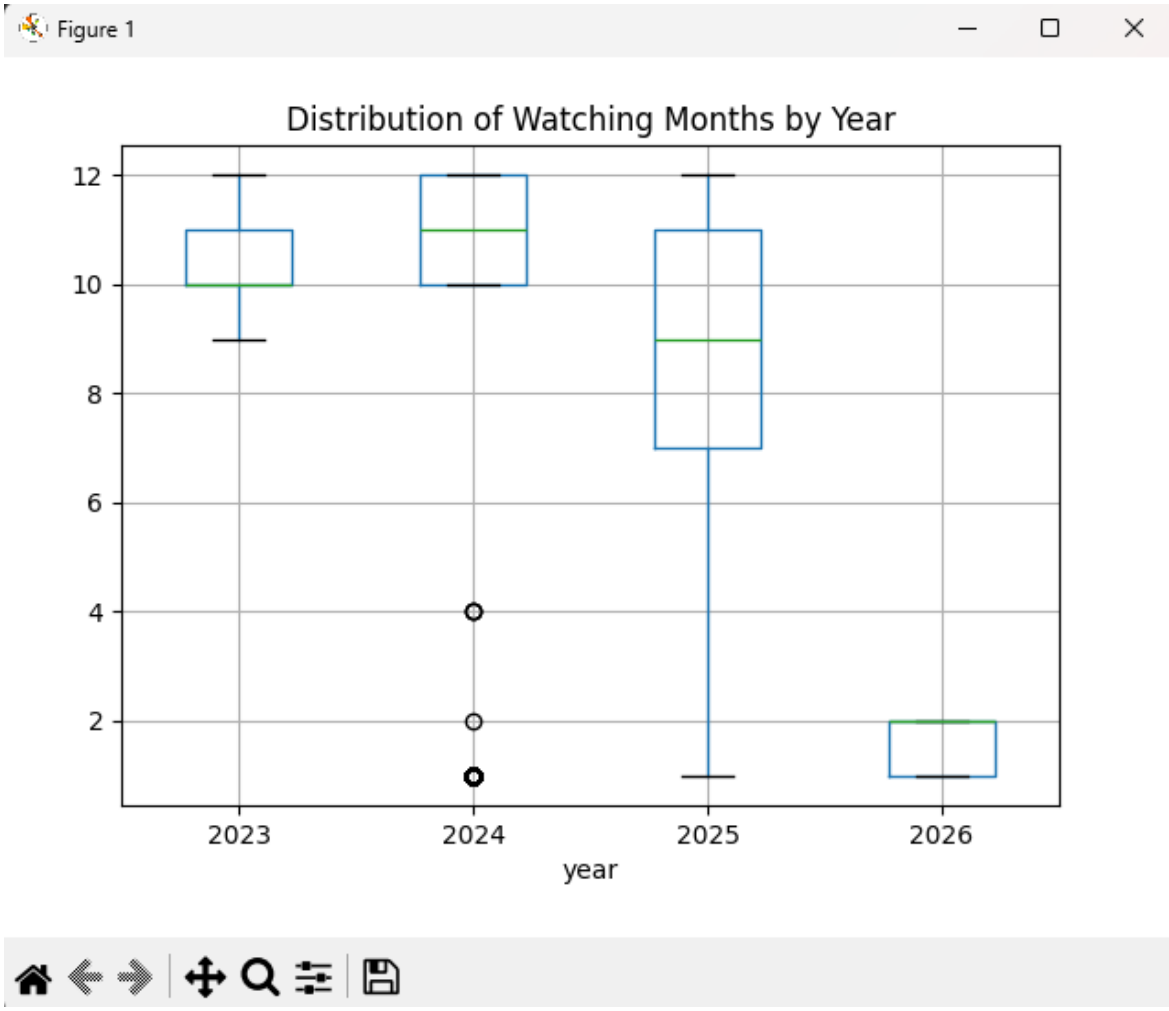
Figure 1: Watch Activity by Year



shows a significant spike in 2025, indicating the highest level of engagement. This suggests that user activity increased substantially during that year. The lower values in 2023 and 2024 indicate either reduced usage or incomplete data collection. The drop in 2026 is expected since it represents only partial-year data and should not be directly compared to a full year.

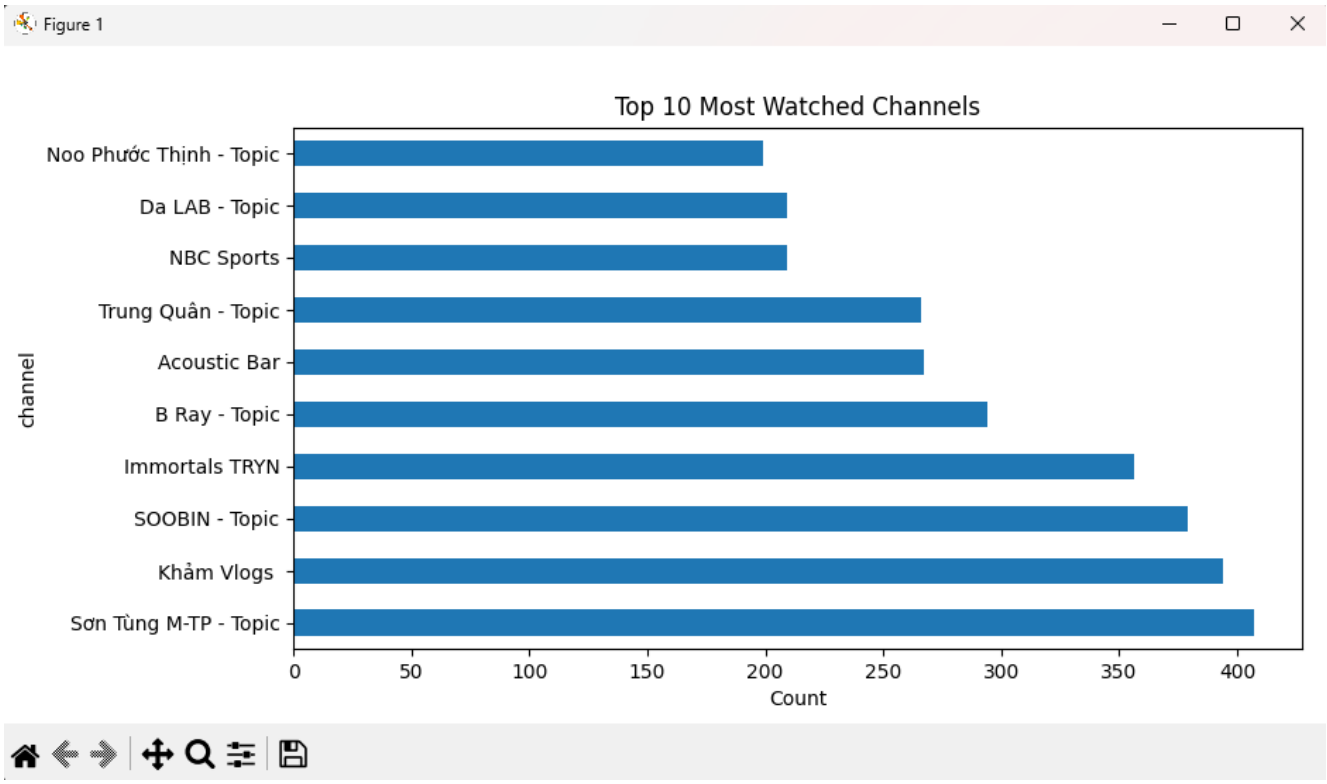
Figure 2: Monthly Trend

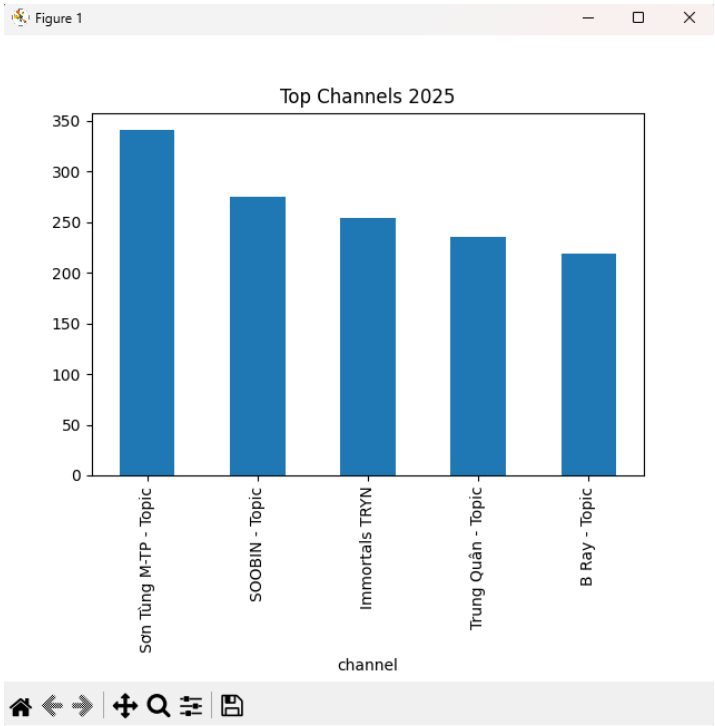


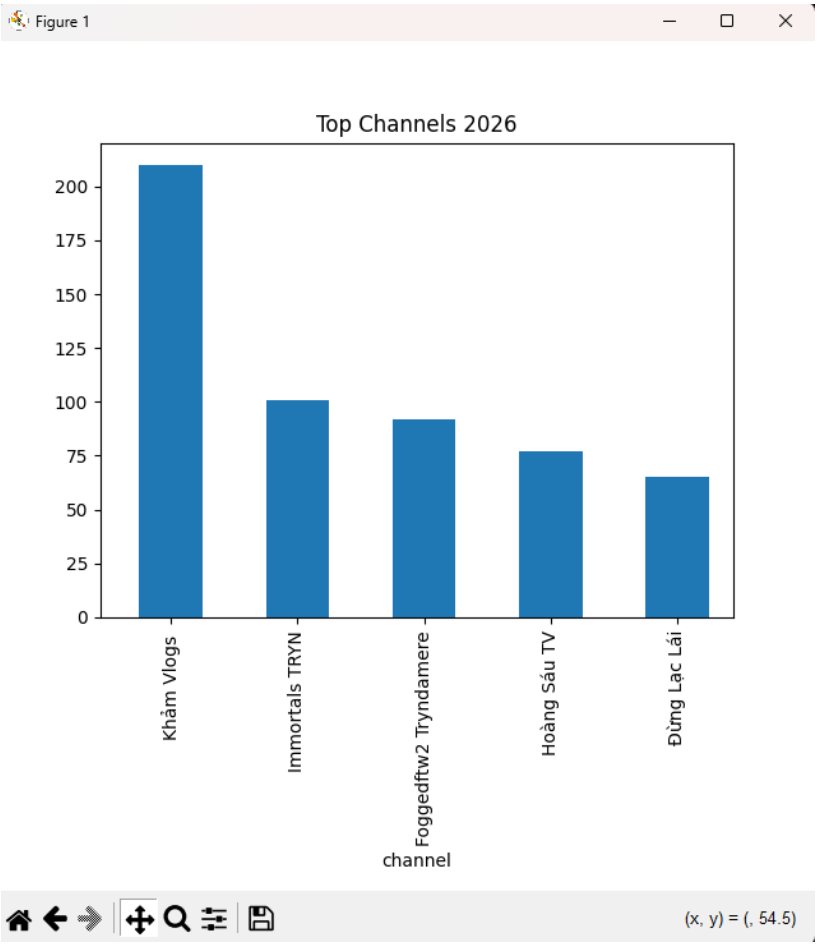


The monthly trend reveals that viewing activity increases during mid-year months, particularly from July to October. This suggests a seasonal pattern where the user is more active during these months. Overall, the trend appears consistent across years, indicating stable viewing habits.

Figure 3: Top Channels

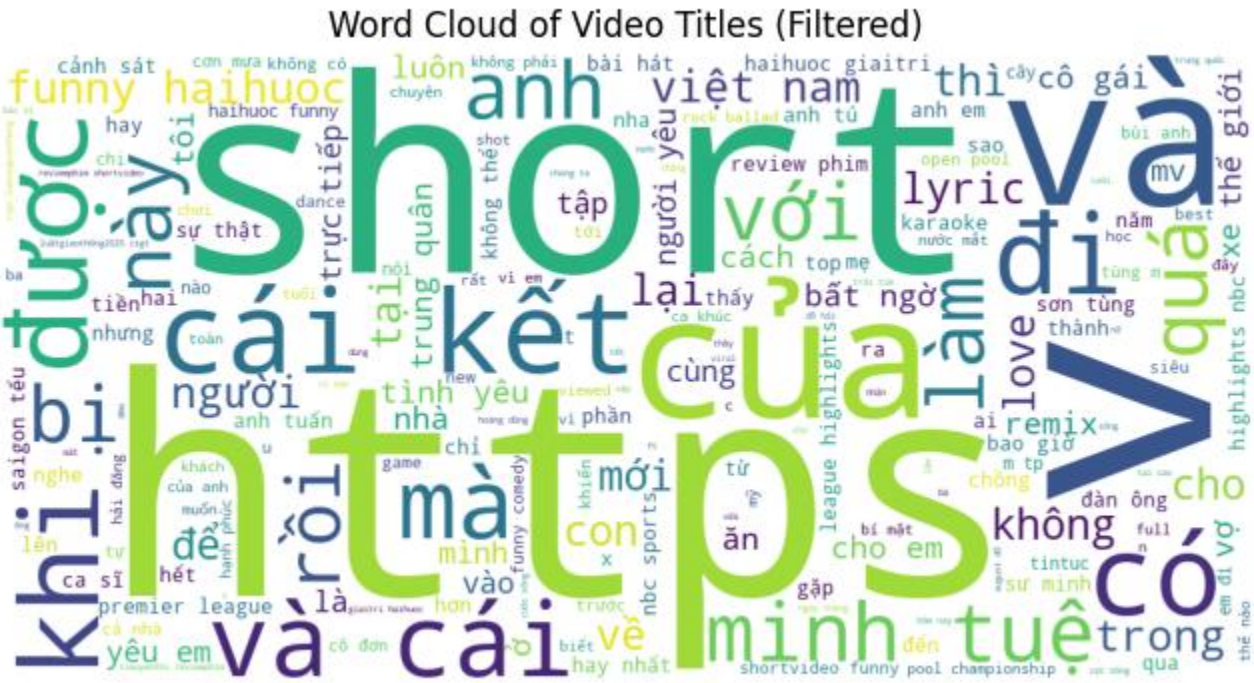






The distribution of top channels demonstrates that the most frequently watched channels are primarily music-related. Many of these channels are Vietnamese artists, indicating a strong preference for music content and repeated engagement with specific creators.

Figure 4: Word Cloud



The word cloud highlights the most frequently occurring words in video titles after removing common filler terms. Prominent words include artist names and music-related keywords such as "reaction," "cover," and "official." This confirms that the dataset is heavily centered around music and entertainment content.

6. Comparative Analysis

Comparison Between Years:

A comparison between the yearly datasets shows that 2025 has significantly higher activity compared to other years. This is likely because it represents a complete year of data, while 2026 only includes partial data.

Despite the difference in volume, the types of content consumed remain consistent across years. The same categories of videos—primarily music and entertainment—continue to dominate. Additionally, the same top channels appear across multiple years, indicating consistent user preferences.

This suggests that while the number of videos watched may vary, the user's interests remain stable over time. Therefore, the difference between years is more quantitative (volume) rather than qualitative (type of content).

7. Key Findings

Deep Insights:

The analysis reveals several deeper insights into user behavior.

First, the dominance of music-related content suggests that the platform is primarily used for music consumption rather than general browsing. The repeated appearance of specific artists indicates strong user loyalty and preference for familiar content.

Second, the seasonal pattern observed in the monthly trend suggests that user engagement is influenced by external factors such as holidays, free time, or lifestyle changes during mid-year periods.

Third, the consistency of top channels across multiple years indicates stable long-term preferences rather than short-term trends. This suggests that the user's taste in content is well-defined and does not fluctuate significantly over time.

Finally, the difference in activity levels between years highlights the importance of data completeness. The lower activity in 2026 should not be interpreted as reduced interest, but rather as incomplete data coverage.

These insights demonstrate that the user exhibits consistent, music-focused behavior with predictable engagement patterns.

8. Conclusion

In conclusion, the exploratory data analysis provides a clear understanding of user behavior in relation to YouTube music content. The results indicate a strong and consistent preference for music and entertainment. While activity levels vary across years, the overall trends remain stable.

This analysis demonstrates the effectiveness of EDA in uncovering meaningful insights from large datasets.

9. AI Usage Disclosure

AI Tool Used: ChatGPT

Prompts Included:
- Perform exploratory data analysis on YouTube dataset

- Generate Python visualization code
- Create report structure and explanations

Evaluation:
ChatGPT was used to assist in structuring the analysis, generating code, and drafting the report. All outputs were reviewed, verified, and customized to match the dataset.

AI Usage Disclosure:

This project utilized ChatGPT (OpenAI GPT-5.3) as an AI assistant to support various stages of the exploratory data analysis.

Prompts Used:

- "Perform exploratory data analysis on YouTube watch history dataset"

- "Explain word cloud and generate code for text visualization"

- "Create a structured report for EDA project"

How the AI Was Used:

ChatGPT was used to assist with:

- Structuring the overall report and organizing sections

- Suggesting appropriate charts such as histograms, line charts, box plots, and word clouds

- Providing explanations for visualizations and interpreting results

- Improving clarity, grammar, and academic tone of the report

Evaluation of AI Performance:

The AI significantly improved efficiency by accelerating code development and report writing. It provided clear guidance on exploratory data analysis techniques and helped identify meaningful insights from the dataset. However, all outputs were reviewed, verified, and modified to ensure correctness and alignment with the actual data. The final analysis and conclusions were validated by the student.

Overall, ChatGPT was an effective tool for enhancing productivity and supporting data analysis, but human verification was essential to ensure accuracy and relevance.